

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical and Instrumentation Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical and Instrumentation Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTEIC701 - PROCESS INSTRUMENTATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which component in a process control loop is responsible for comparing the measured value with the setpoint?
 - a) Final control element
 - b) Controller
 - c) Process
 - d) Sensor
 - 2 What is the primary purpose of Advanced Process Instrumentation in modern industries?
 - a) Reduce the number of sensors
 - b) Improve accuracy, reliability, and control efficiency
 - c) Increase manual operations
 - d) Eliminate process automation
 - 3 In a distillation column, the main purpose of reflux control is to:
 - a) Maintain bottom product temperature
 - b) Improve separation by returning part of the condensed vapor to the column
 - c) Increase feed flow rate
 - d) Reduce column pressure
 - 4 Flow control of distillate and bottoms is typically achieved using
 - a) Pressure indicators only
 - b) Control valves placed on product outlet lines
 - c) Manual adjustments by the operator
 - d) Level switches in the reflux drum
 - 5 In a boiler control system, which of the following is typically a manipulated variable?
 - a) Steam pressure
 - b) Fuel flow rate
 - c) Drum level
 - d) Steam temperature

- 6 Air-to-fuel ratio control is necessary mainly to:
 - a) Increase boiler drum size
 - b) Maintain proper combustion and reduce emissions
 - c) Reduce feedwater flow
 - d) Lower furnace draft completely
- 7 In compressors, capacity control is commonly achieved using:
 - a) Changing compressor material
 - b) Inlet guide vanes or variable speed drive
 - c) Adjusting lubrication oil level
 - d) Installing a larger intercooler
- 8 In pump control systems, a common manipulated variable is
 - a) Suction pressure
 - b) Discharge valve position
 - c) Pump casing temperature
 - d) Pump impeller diameter
- 9 Risk in process safety is generally defined as:
 - a) Probability of failure only
 - b) Consequence of an event only
 - c) Probability $\times$ Consequence of an undesired event
 - d) A measure of equipment cost
- 10 Process Hazard Analysis (PHA) is mainly used to:
 - a) Improve plant aesthetics
 - b) Identify and analyze potential hazards in processes
 - c) Increase production rate
 - d) Select raw materials only
- 11 HAZOP study is based on:
 - a) Energy balance equations
 - b) Guidewords applied to process parameters
 - c) Control loop tuning
 - d) Cost optimization
- 12 Direct Digital Control (DDC) primarily means:
 - a) All control actions are mechanical
 - b) Control is performed by analog computers
 - c) Controller logic is implemented using digital computers
 - d) No sensor is used in the loop

Q2. Solve the following.

- A) Explain the components of a process control loop with a neat block diagram. 6
- B) Describe the concept and need of Advanced Process Instrumentation in modern industries. Explain how it improves accuracy, reliability, and automation. 6

Q3. Solve the following.

- A) What is reflux control in distillation? Explain different reflux control strategies and their impact on separation efficiency and product purity. 6
- B) Explain pressure and temperature control methods used in distillation columns. 6

Q4. Solve Any Two of the following.

- A) Explain the overall operation of a boiler with a neat diagram. 6
- B) Describe safety interlocks and the Burner Management System (BMS). 6

- C) Describe steam temperature control in boilers in detail. 6
- Q5. Solve Any Two of the following.
- A) Explain the types and operation of compressors (centrifugal, rotary, and reciprocating). 6
- B) What is surge in centrifugal compressors? Explain surge control system with suitable diagram. 6
- C) Describe the role of sensors, transmitters, controllers, and control valves in ensuring safe and efficient operation of compressors in industrial systems. 6
- Q6. Solve Any Two of the following.
- A) Explain the concept of process safety. Discuss risk, consequence, and major risk terminologies used in industrial safety management. 6
- B) Describe Process Hazard Analysis (PHA). Explain major PHA techniques. 6
- C) What is a HAZOP study? Explain its methodology, guidewords, team structure, and advantages in identifying process deviations. 6

***** End *****